① MST (optimisation version)

INPUT: A WEIGHTED GRAPH G

OUTPUT: A SPANNING TREE OF G OF MINIMUM WEIGHT

MST (decision version)

INPUT: A WEIGHTED GRAPH G & A THRESHOLD T

OUTPUT: YES IF G HAS A SPANNING TREE OF WEIGHT $\leq T$

NO: OTHERWISE

$S_1 = ABB$
$S_2 = BBA$

ABBBBA

ABBA

$$\begin{array}{ccc} A & \boxed{B\ B} & \\ & \boxed{B\ B} & A \end{array}$$

Conjecture: $SCS(S_1, S_2) = |S_1| + |S_2| - LCS(S_1, S_2)$

SCS (decision version)

INPUT: STRING S_1, S_2 INTEGER K

OUTPUT: YES IF $\exists$ STRING T w/ $|T| \leq K$ s.t. S_1, S_2 ARE ~~SUBSTRINGS~~ SUBSEQUENCES OF T

NO OTHERWISE

(2) REDUCTION SCS → LCS

Given instance s_1, s_2, k of SCS

$f(s_1, s_2, k) = (s_1, s_2, |s_1| + |s_2| - k)$

~~MAS ≤ k ⇔ |s_1| + |s_2| - LCS ≥ |s_1| + |s_2| - k~~

$|SCS \le k \Leftrightarrow |s_1| + |s_2| - LCS \le k$

$\Leftrightarrow LCS \geq |s_1| + |s_2| - k)$

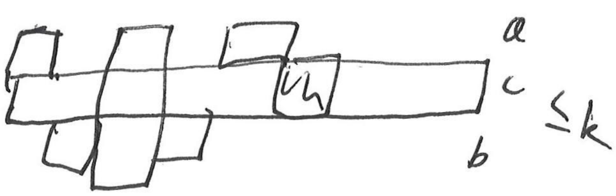
Property 1: f is poly-time computable ✓

Property 2: $\ell \in SCS \Leftrightarrow f(\ell) \in LCS$

⇑

$\exists t$ s.t. $|t| \le k$,

$s_1, s_2 \le t$



$a+b+c \le k$

$a+c = |s_1|$

$b+c = |s_2|$

$c \le LCS(s_1, s_2)$

$a+b+c \geq |s_1| + |s_2| - LCS(s_1, s_2) \le k$